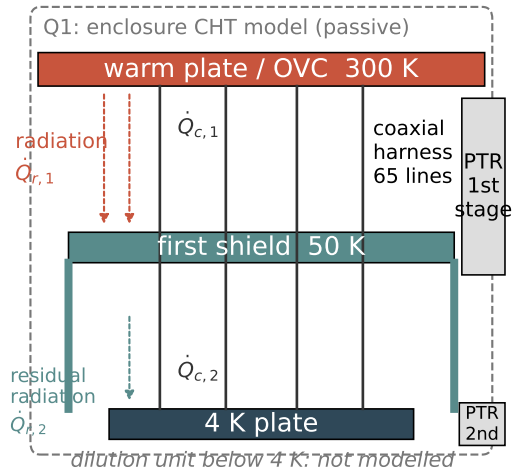
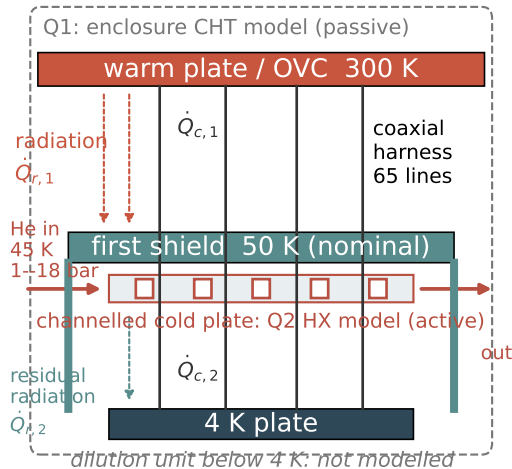


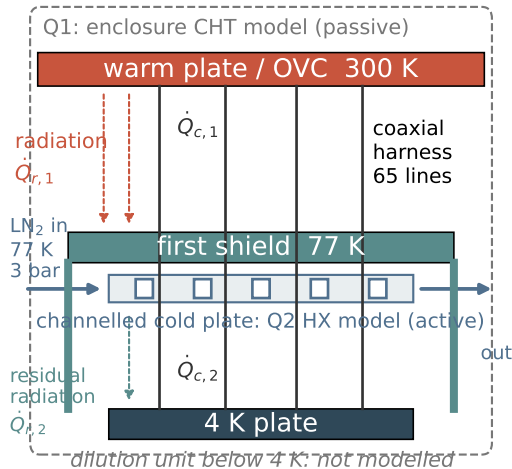
(a) reference: PTR first stage
40--50 K, conductive



(b) route He: circulated gas
45 K in, 1--18 bar



(c) route N₂: liquid nitrogen
77 K in, 3 bar



first-stage duty removed by conduction to the PTR;
no coolant circuit (used for the reference wall temperature only)

duty $\dot{Q}_1$ removed by the coolant:
He cryoplant and circulator

duty $\dot{Q}_1$ removed by the coolant:
LN₂ dewar or pressurised loop